

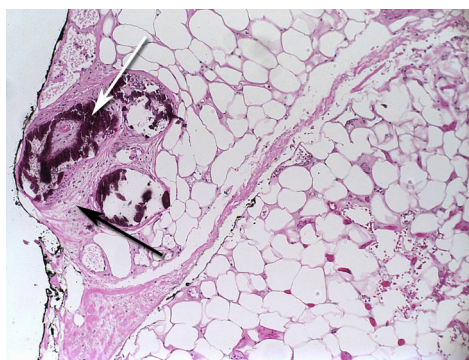


Fig 2. Nonuremic calciphylaxis in a patient treated with teriparatide: intramural calcium deposition (*white arrow*) in subcutaneous arterioles with intimal hyperplasia (*black arrow*) and ischemic changes of the surrounding tissue. (Hematoxylin-eosin stain; original magnification: $\times 100$.)

intractable pain despite management by pain specialists. Ultimately, she was transitioned to hospice care and died soon thereafter.

This case underscores the importance of high clinical suspicion and prompt diagnosis in patients presenting with livedoid erythema, painful indurated subcutaneous plaques, and ulcerations in adipose-rich areas who have risk factors for nonuremic calciphylaxis including female gender, obesity, chronic inflammation, liver disease, rapid weight loss, malignancy, and primary hyperparathyroidism.¹ Corticosteroids,¹ warfarin,¹ and teriparatide² have all been implicated as well. These factors are all thought to disrupt the balance between nuclear factor-kappa beta (NF κ B)-mediated activation and fetuin-A and matrix Gla-protein-mediated (MGP) inhibition of endothelial cell calcification.³ Because of the patient's multiple risk factors for calciphylaxis, the relationship to teriparatide cannot be stated with certainty, however previous cases exist.^{2,4}

Calciphylaxis may be diagnosed clinically in patients with typical skin findings and multiple risk factors. Because histopathology is often nondiagnostic in calciphylaxis, the benefits of a biopsy must be weighed against the risk of poor wound healing. If biopsy is performed, special stains may aid in diagnosis as histopathologic changes of early calciphylaxis are subtle.⁵ Clearer definition of diagnostic criteria for calciphylaxis is needed.

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Alcohol-induced vasculitis: Case report and commentary

To the Editor: A 26-year-old woman with no significant medical history on no medications or supplements was evaluated for a 2-year history of episodic tender, nonpruritic, purpuric macules and papules on the back of her feet and legs with associated edema. She reported that several hours after drinking beer, wine, or liquor, she develops purpuric lesions (Fig 1) that resolved within 2 to 7 days without sequelae. She had experienced at least 8 episodes, each directly linked to alcohol ingestion. She had no skin lesions in the 6 months before evaluation during which she abstained from drinking any alcohol.

A provocation test consisted of drinking 8 oz of alcohol. Within 6 hours she developed subtle pink macules and papules on her legs. Eight hours after ingestion, 1 of these lesions was biopsied and revealed a perivascular neutrophilic infiltrate and leukocytoclasia with focal vascular wall damage and fibrinoid changes (Fig 2). On direct immunofluorescence, 1 small superficial dermal vessel demonstrated granular perivascular staining with IgA (not shown).

Four cases of alcohol-associated purpura have been reported in the literature¹⁻⁴; none of these cases included biopsy evidence of vascular wall damage. Alibrandi et al¹ reported a case of a 60-year-old man with recurrent asymptomatic



Fig 1. Vasculitis. Clinical photograph of patient's left ankle and foot showing petechiae.

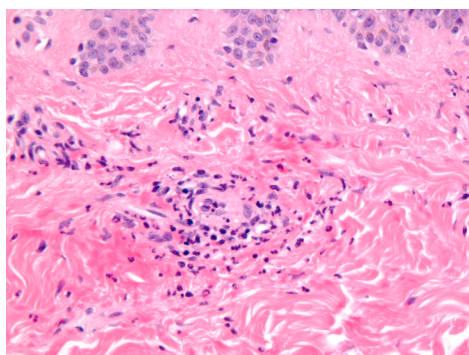


Fig 2. Vasculitis. Punch skin biopsy specimen demonstrating a dermal blood vessel with fibrinoid changes in the context of neutrophilic inflammation and leukocytoclasia. (Hematoxylin-eosin stain; original magnification: $\times 400$.)

nonpalpable purpuric macules on the legs after ingesting vinegar or alcohol. Biopsy specimen of a lesion 3 hours after drinking 250 mL of 10% ethanol did not show signs of vasculitis, but direct immunofluorescence revealed IgA deposition in the dermal vessels. Chua et al³ reported a case of a 27-year-old man with recurrent palpable purpura on the ankles after drinking 2 pints of beer. Biopsy specimen of lesions 1 and 2 hours after consumption of 400 mL of 10% sterile alcohol showed a perivascular lymphocytic infiltrate in the papillary dermis with positive IgA staining on direct immunofluorescence and fibrin staining on immunofluorescence of both lesional and nonlesional skin. A case of Sjögren's syndrome associated with alcohol-induced purpura, and a case of purpura considered as a result of aspirin-induced platelet dysfunction aggravated by drinking alcohol have been reported; however in these cases skin biopsies were not performed.^{2,4}

The mechanism by which alcohol may induce purpura is unclear. The current case suggests a cause

related to an immune-complex vasculitis mediated by IgA. Alibrandi et al¹ speculated that degranulation of mast cells may cause vasodilation allowing for immune complexes to adhere to blood-vessel basal membranes. However, Chua et al³ did not find evidence of elevated serum tryptase levels after alcohol rechallenge.

Drug-induced vasculitis can be defined as any type of inflammatory vascular disorder caused by administration of a pharmacologic agent.⁵ Dermatologic changes are among the most common clinical manifestations and are often characterized by symmetric purpuric macules or papules on the bilateral lower extremities. The exclusion of other possible causative agents by patient history, the clinical pattern, a positive rechallenge, and the biopsy results all provide strong evidence that this is a case of alcohol-induced cutaneous vasculitis.

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